SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: Hand Gesture Virtual Mouse Web Application

1. Introduction

This document specifies the requirements for the Hand Gesture Virtual Mouse Web Application. The system enables users to control mouse movements using hand gestures captured via webcam.

2. Purpose

The purpose of this project is to provide a gesture-based alternative to traditional mouse devices using computer vision techniques. It is intended for academic use, AI learning, and Human Computer Interaction research.

3. Scope

The system performs real-time hand detection, landmark extraction, gesture recognition, cursor movement, and click execution via a web interface.

4. Definitions

- *HCI* – Human Computer Interaction
- *SRS* – Software Requirement Specification
- *FPS* – Frames Per Second

5. Overall Description

The application captures webcam input from the browser, sends frames to Flask server, processes them using MediaPipe, identifies gestures, and maps them to mouse actions.

Architecture:
Webcam → Flask Server → MediaPipe → Gesture Logic → Mouse Controller → Browser Output

6. User Classes

- Students
- Developers
- Faculty / Evaluators

7. Operating Environment

- Windows/Linux OS
- Python 3.x
- Chrome/Edge Browser
- Internet Connection

8. Functional Requirements

- **FR1:** System shall access webcam.
- **FR2:** System shall detect hands.
- **FR3:** System shall extract 21 landmarks.
- **FR4:** Index finger controls cursor.
- **FR5:** Thumb + Index pinch performs click.
- **FR6:** Video stream shown in browser.
- **FR7:** Application deployed on Render.

9. Non-Functional Requirements

- **Performance:** Minimum 15 FPS.
- **Latency** below 200ms.
- **Usability:** Simple UI.
- **Reliability:** Camera failure recovery.
- **Security:** No video storage.
- **Portability:** Runs on any modern browser.

10.Hardware Requirements

- Webcam
- 4GB RAM minimum
- Stable Internet

11.Software Requirements

- Python 3.x
- Flask
- MediaPipe
- OpenCV
- PyAutoGUI
- Render Account

12.Constraints

- Lighting affects accuracy.
- Render free tier limits background tasks.
- Browser permission required.

13.Future Enhancements

- Right click gesture
- Scroll gesture
- Multi hand support
- Mobile version
- Gesture customization

14.Conclusion

The Hand Gesture Virtual Mouse Web Application demonstrates real-time computer vision, gesture recognition, and cloud deployment. It serves as a strong academic project showcasing AI and HCI integration